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\\MARTINOS DEVELOPER

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\\MARTINOS DEVELOPER\HUBER\Physics_playground\Executed_20260603_Skope\localizer

TA: 15 sec Coil Selection: Manual Voxel Size: 0.5×0.5×5.0 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	On
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	All Segments
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	8.6 ms
TE	3.69 ms
Averages	2
Concatenations	3
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	8.6 ms
TE	3.69 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None

Contrast - Common

Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	256
Phase Resolution	100 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve	Off
Phase Partial Fourier	Off
Asymmetric Echo	Allowed

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	1
Distance Factor	20 %
Position	Isocenter

Geometry - Common

Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	8.6 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	3

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up

System - Adjustments

B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.

System - Tx/Rx

Frequency 1H	297.135464 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	8.6 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	250 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Sequence - Assistant

Allowed Delay	0 s
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Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	20 deg
Measurements	1
Contrasts	1
TE	3.69 ms
TR	8.6 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	qfl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	320 Hz/Px
Asymmetric Echo	Allowed
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Acoustic noise reduction	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz
Coil Elements	AC

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\Physics_playground\Executed_20260603_Skopelwip_ShimToolbox_B0
map_3mm

TA: 31 sec Coil Selection: Manual Voxel Size: 3.0x3.0x3.0 mm³ Acc:: 2 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.8 A21.6 F24.1 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	52
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
TR	15.0 ms
TE 1	2.20 ms
TE 2	5.40 ms
Concatenations	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	15.0 ms
TE 1	2.20 ms
TE 2	5.40 ms
MTC	Off
Magn. Preparation	None
Flip Angle	8 deg
Fat-Water Contrast	Water Excitation
Dark Blood	Off
Contrasts	2
SWI	Off
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
Base Resolution	72
Phase Resolution	100 %
Slice Resolution	100 %

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.8 A21.6 F24.1 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	52
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
TR	15.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.8 A21.6 F24.1 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.8 A21.6 F24.1
L	0.8 mm
A	21.6 mm
F	24.1 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.8 A21.6 F24.1 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	216 mm
F >> H	216 mm
R >> L	156 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.135464 MHz
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System - Tx/Rx

? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	15.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Water Excitation
Magn. Preparation	None
Dark Blood	Off
FOV Read	216 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing

Inline - MapIt

MapIt	None
Flip Angle	8 deg
Measurements	1
Contrasts	2
TE 1	2.20 ms
TE 2	5.40 ms
TR	15.0 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	WIP_GRE
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Readout Mode	Monopolar
Gradient Mode	Whisper
Flow Compensation 1	None
Flow Compensation 2	None
Bandwidth 1	500 Hz/Px
Bandwidth 2	500 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Acoustic noise reduction	Off

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz
Coil Elements	AC

Sequence - Special

Mode	B0 Mapping
Masking	Brain Extraction
Phase Unwrapping	Spatial

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\MARTINOS DEVELOPER\HUBER\Physics_playground\Executed_20260603_Skopelcalib

TA: 5 sec Coil Selection: Manual Voxel Size: 1.6×1.6×200.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	1
Slice Oversampling	0.0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	200.000 mm
TR	100.000 ms
TE	10.00 ms
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	100.000 ms
TE	10.00 ms
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off

Resolution - Common

FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	200.000 mm
Base Resolution	128
Phase Resolution	100 %

Resolution - Common

Slice Resolution	50 %
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Resolution - Acceleration

Acceleration Mode	None
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	1
Slice Oversampling	0.0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	200.000 mm
TR	100.000 ms
Series	Ascending

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T

System - Miscellaneous

Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	On

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
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System - Tx/Rx

Frequency 1H	297.135464 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	100.000 ms

Sequence - Part 1

Sequence Name	opc
Dimension	3D
Gradient Mode	Fast

Sequence - Part 2**Sequence - Nuclei**

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz
Coil Elements	AC

Sequence - Special

Pulseseq file	Saved/Haotian/Skope/skope_offresAndPosCalib_ext1.seq
Execution mode	pulseseq/run

Sequence - Special

Data handling	none
Timing and Flip Angles	adapt FA
FOV positioning	enabled
Orientation mapping	XYZ in TRA
libBalance / Grad health	interpolate
Gradient switching frequency	0 Hz
Number of runs	1
Delay after run(s)	0.00 ms
ADC length per segment	0
Use SET for SLC	Off
Delay before start	0 ms
Gradient scaling[x]	100.00 %
Gradient scaling[y]	100.00 %
Gradient scaling[z]	100.00 %
Overall RF scaling	100.00 %
RF scaling [exc]	100.00 %
RF scaling [ref]	100.00 %
RF scaling other	100.00 %

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\Physics_playground\Executed_20260603_Skopellarge_fov_no_third_order_short_termECC

TA: 15 sec Coil Selection: Manual Voxel Size: 1.0x1.0x1.0 mm³ Acc:: 4 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	18
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	51.5 ms
Vol. TR	927 ms
TE 1	19.00 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	51.5 ms
Vol. TR	927 ms
TE 1	19.00 ms
Multi-echo spacing	40.20 ms
MTC	Off
Flip Angle	18 deg
Fat-Water Contrast	Fast Fat Saturation
Magn. Prep. Shots	1
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
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Contrast - Dynamic

Measurements	10
Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	206
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	24
Acceleration Factor 3D	1
Reference Lines 3D	18
Reordering Shift 3D	0
Phase Partial Fourier	6/8
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	18
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	51.5 ms
Vol. TR	927 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 H0.0 mm

Geometry - AutoAlign

Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 H0.0
L	0.0 mm
A	20.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Quick
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.6 A4.0 H0.6 mm
! Orientation	T > S0.4 > C-0.1
! Rotation	90.14 deg
! R >> L	123 mm
! A >> P	169 mm
! F >> H	42 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.135464 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00

System - Tx/Rx

Image Scaling	0.100
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Sequence - Part 1

Sequence Name	vaso a2d6453
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1104 Hz/Px
Echo Spacing	1.01 ms
Turbo Factor	18
Segmentation	1
EPI Factor	39

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	1000 us
RF time x BW	15
PAT ref. FA	3 deg
Fat sat. FA	110 deg
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Ramp Sampling	On
Trigger per shot	On
Noise image	Off
Relax spoilers	On
Round up Vol. TR	Off
MT flip angle	500 deg
MT off-res.	2000 Hz
MT RF duration	10240 us
Custom Water Exc.	-none-
Phase Correction	integrated
Saturation RF	per Shot
EPI rise time factor	1.20
G. spoil dephasing[1]	0.0 pi
G. spoil dephasing[2]	4.0 pi
G. spoil dephasing[3]	2.0 pi
Modify Ice Config	On
G-factor map	Off
GRAPPA Regularization	100 /10^6
Slab Scale	-10 %
RF spoil scheme	Conventional
Read polarity	Dual-polarity

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\Physics_playground\Executed_20260603_Skopellarge_fov_no_third_order_short_notermECC

TA: 15 sec Coil Selection: Manual Voxel Size: 1.0x1.0x1.0 mm³ Acc:: 4 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	18
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	51.5 ms
Vol. TR	927 ms
TE 1	19.00 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	51.5 ms
Vol. TR	927 ms
TE 1	19.00 ms
Multi-echo spacing	40.20 ms
MTC	Off
Flip Angle	18 deg
Fat-Water Contrast	Fast Fat Saturation
Magn. Prep. Shots	1
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
--------------	----------

Contrast - Dynamic

Measurements	10
Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	206
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	24
Acceleration Factor 3D	1
Reference Lines 3D	18
Reordering Shift 3D	0
Phase Partial Fourier	6/8
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	18
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	51.5 ms
Vol. TR	927 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 H0.0 mm

Geometry - AutoAlign

Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 H0.0
L	0.0 mm
A	20.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Quick
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.6 A4.0 H0.6 mm
! Orientation	T > S0.4 > C-0.1
! Rotation	90.14 deg
! R >> L	123 mm
! A >> P	169 mm
! F >> H	42 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.135464 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00

System - Tx/Rx

Image Scaling	0.100
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Sequence - Part 1

Sequence Name	vaso a2d6453
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1104 Hz/Px
Echo Spacing	1.01 ms
Turbo Factor	18
Segmentation	1
EPI Factor	39

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	1000 us
RF time x BW	15
PAT ref. FA	3 deg
Fat sat. FA	110 deg
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Ramp Sampling	On
Trigger per shot	On
Noise image	Off
Relax spoilers	On
Round up Vol. TR	Off
MT flip angle	500 deg
MT off-res.	2000 Hz
MT RF duration	10240 us
Custom Water Exc.	-none-
Phase Correction	integrated
Saturation RF	per Shot
EPI rise time factor	1.20
G. spoil dephasing[1]	0.0 pi
G. spoil dephasing[2]	4.0 pi
G. spoil dephasing[3]	2.0 pi
Modify Ice Config	On
G-factor map	Off
GRAPPA Regularization	100 /10^6
Slab Scale	-10 %
RF spoil scheme	Conventional
Read polarity	Dual-polarity

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\Physics_playground\Executed_20260603_Skopelvaso_inv_long_inversion

TA: 1:16 min Coil Selection: Manual Voxel Size: 1.0x1.0x1.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	18
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	51.5 ms
Vol. TR	7439.2 ms
TE 1	19.00 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	51.5 ms
Vol. TR	7439.2 ms
TE 1	19.00 ms
Multi-echo spacing	40.20 ms
MTC	Off
Magn. Preparation	Non-sel. IR
TI 1	1917.10 ms
TI 2	5625.10 ms
Flip Angle	50 deg
Fat-Water Contrast	Fast Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	10
Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	206
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	18
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	51.5 ms
Vol. TR	7439.2 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 H0.0
L	0.0 mm

Geometry - AutoAlign

A	20.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Quick
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.6 A4.0 H0.6 mm
! Orientation	T > S0.4 > C-0.1
! Rotation	90.14 deg
! R >> L	123 mm
! A >> P	169 mm
! F >> H	42 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.135464 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	0.200

Sequence - Part 1

Sequence Name	vso vaso_v6-dev
Dimension	3D

Sequence - Part 1

Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1104 Hz/Px
Echo Spacing	1.01 ms
Turbo Factor	72
Segmentation	4
EPI Factor	39

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	1000 us
RF time x BW	15
PAT ref. FA	3 deg
Fat sat. FA	110 deg
Variable FA	4
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Ramp Sampling	On
Trigger per shot	On
Noise image	Off
Relax spoilers	Off
Round up Vol. TR	Off
MT flip angle	500 deg
MT off-res.	2000 Hz
MT RF duration	10240 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI rise time factor	1.20
G. spoil dephasing[1]	0.0 pi
G. spoil dephasing[2]	4.0 pi
G. spoil dephasing[3]	2.0 pi
Modify Ice Config	On
G-factor map	Off
GRAPPA Regularization	10 /10^6
Slab Scale	-10 %
RF spoil scheme	Conventional
Read polarity	Inverted RO
Pre-T11 delay	0.00 ms
Pre-T12 delay	0.00 ms
Pre-Inv delay	0.00 ms

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\Physics_playground\Executed_20260603_Skope\wip_3DEPI_disconn_ withstECC

TA: 16 sec Coil Selection: Manual Voxel Size: 0.9×0.9×0.9 mm³ Acc:: 3 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	-50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	30
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	0.88 mm
TR	61.0 ms
TE	25.00 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	61.0 ms
TE	25.00 ms
MTC	Off
Flip Angle	7 deg
Fat-Water Contrast	Fat Saturation
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	5
Multiple Series	Off
3D Reordering	Standard

Contrast - Dynamic

Reordering	Linear
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Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	0.88 mm
Base Resolution	224
Phase Resolution	98 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Reference Scans	GRE/Separate
Acceleration Factor PE	1
Reference Lines PE	24
Acceleration Factor 3D	3
Reference Lines 3D	24
Reordering Shift 3D	1
Deep Resolve	Off
Phase Partial Fourier	Off
Slice Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	30
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	0.88 mm
TR	61.0 ms
Multi-Slice Mode	Sequential
Series	Ascending

Geometry - Common

Concatenations	1
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Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	Isocenter
! Orientation	Transversal
! Rotation	0.00 deg
! A >> P	263 mm
! R >> L	350 mm
! F >> H	69 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.135464 MHz
I Ref. Amplitude 1H	370.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	61.0 ms
Segments	2
Concatenations	1

Sequence - Part 1

Sequence Name	WIP_epi
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast*
Flow Compensation	None
Reordering	Linear
Bandwidth	1174 Hz/Px
Echo Spacing	1.00 ms
Free Echo Spacing	Off
Segments	2
EPI Factor	39

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Correction	Internal

Sequence - Special

Prep/dummy scan duration	0 sec
Selective RF Duration	1024 us
Desired RF TBW	8.0
Flyback	No
QuickSat	1
Excitation Orientation	SS - Slice Selective
Ernst Angle T1	1220 ms
Reduce Slab Thickness?	Weak
skipped-CAIPI?	Off
Log Signals	Off

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\Physics_playground\Executed_20260603_Skope\wip_3DEPI_disconn_
wostECC

TA: 16 sec Coil Selection: Manual Voxel Size: 0.9×0.9×0.9 mm³ Acc:: 3 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	-50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	30
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	0.88 mm
TR	61.0 ms
TE	25.00 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	61.0 ms
TE	25.00 ms
MTC	Off
Flip Angle	7 deg
Fat-Water Contrast	Fat Saturation
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	5
Multiple Series	Off
3D Reordering	Standard

Contrast - Dynamic

Reordering	Linear
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Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	0.88 mm
Base Resolution	224
Phase Resolution	98 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Reference Scans	GRE/Separate
Acceleration Factor PE	1
Reference Lines PE	24
Acceleration Factor 3D	3
Reference Lines 3D	24
Reordering Shift 3D	1
Deep Resolve	Off
Phase Partial Fourier	Off
Slice Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	30
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	0.88 mm
TR	61.0 ms
Multi-Slice Mode	Sequential
Series	Ascending

Geometry - Common

Concatenations	1
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Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	Isocenter
! Orientation	Transversal
! Rotation	0.00 deg
! A >> P	263 mm
! R >> L	350 mm
! F >> H	69 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.135464 MHz
I Ref. Amplitude 1H	370.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	61.0 ms
Segments	2
Concatenations	1

Sequence - Part 1

Sequence Name	WIP_epi
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast*
Flow Compensation	None
Reordering	Linear
Bandwidth	1174 Hz/Px
Echo Spacing	1.00 ms
Free Echo Spacing	Off
Segments	2
EPI Factor	39

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Correction	Internal

Sequence - Special

Prep/dummy scan duration	0 sec
Selective RF Duration	1024 us
Desired RF TBW	8.0
Flyback	No
QuickSat	1
Excitation Orientation	SS - Slice Selective
Ernst Angle T1	1220 ms
Reduce Slab Thickness?	Weak
skipped-CAIPI?	Off
Log Signals	Off

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\Physics_playground\Executed_20260603_Skopel\vaso_2p3mm

TA: 43 sec Coil Selection: Manual Voxel Size: 2.3×2.3×2.3 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	40
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.30 mm
TR	47.4 ms
Vol. TR	4115.2 ms
TE 1	15.70 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	47.4 ms
Vol. TR	4115.2 ms
TE 1	15.70 ms
Multi-echo spacing	42.40 ms
MTC	Off
Magn. Preparation	Non-sel. IR
TI 1	1259.60 ms
TI 2	3155.60 ms
Flip Angle	18 deg
Fat-Water Contrast	Fast Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
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Contrast - Dynamic

Measurements	10
Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.30 mm
Base Resolution	92
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	40
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.30 mm
TR	47.4 ms
Vol. TR	4115.2 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 H0.0
L	0.0 mm
A	20.0 mm
H	0.0 mm

Geometry - AutoAlign

Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Quick
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.6 A6.7 F0.3 mm
! Orientation	T > S0.4 > C-0.1
! Rotation	90.14 deg
! R >> L	123 mm
! A >> P	169 mm
! F >> H	107 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.135464 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	0.200

Sequence - Part 1

Sequence Name	vso vaso_v6-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal

Sequence - Part 1

Gradient Mode	Fast
Reordering	Linear
Bandwidth	1874 Hz/Px
Echo Spacing	0.60 ms
Turbo Factor	40
Segmentation	1
EPI Factor	70

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	1000 us
RF time x BW	15
PAT ref. FA	3 deg
Fat sat. FA	110 deg
Variable FA	4
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Ramp Sampling	On
Trigger per shot	On
Noise image	Off
Relax spoilers	Off
Round up Vol. TR	Off
MT flip angle	500 deg
MT off-res.	2000 Hz
MT RF duration	10240 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI rise time factor	1.20
G. spoil dephasing[1]	0.0 pi
G. spoil dephasing[2]	4.0 pi
G. spoil dephasing[3]	2.0 pi
Modify Ice Config	On
G-factor map	Off
GRAPPA Regularization	10 / 10^6
Slab Scale	-10 %
RF spoil scheme	Conventional
Read polarity	Inverted RO
Pre-T11 delay	300.00 ms
Pre-T12 delay	0.00 ms
Pre-Inv delay	0.00 ms

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\Physics_playground\Executed_20260603_Skopelwip_3DEPI_seg_fid_E
S1

TA: 7 sec Coil Selection: Manual Voxel Size: 0.9×0.9×0.9 mm³ Acc:: 3 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	-50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	30
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	190 mm
FOV Phase	100.0 %
Slice Thickness	0.88 mm
TR	50.0 ms
TE	19.00 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	50.0 ms
TE	19.00 ms
MTC	Off
Flip Angle	10 deg
Fat-Water Contrast	Fat Saturation
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off
3D Reordering	Standard

Contrast - Dynamic

Reordering	Linear
------------	--------

Resolution - Common

FOV Read	190 mm
FOV Phase	100.0 %
Slice Thickness	0.88 mm
Base Resolution	208
Phase Resolution	98 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Reference Scans	GRE/Separate
Acceleration Factor PE	1
Reference Lines PE	24
Acceleration Factor 3D	3
Reference Lines 3D	24
Reordering Shift 3D	1
Deep Resolve	Off
Phase Partial Fourier	Off
Slice Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	30
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	190 mm
FOV Phase	100.0 %
Slice Thickness	0.88 mm
TR	50.0 ms
Multi-Slice Mode	Sequential
Series	Ascending

Geometry - Common

Concatenations	1
----------------	---

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.135464 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	50.0 ms
Segments	3
Concatenations	1

Sequence - Part 1

Sequence Name	WIP_epi
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast*
Flow Compensation	None
Reordering	Linear
Bandwidth	1202 Hz/Px
Echo Spacing	1.00 ms
Free Echo Spacing	Off
Segments	3
EPI Factor	25

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Correction	Internal

Sequence - Special

Prep/dummy scan duration	2 sec
Selective RF Duration	1024 us
Desired RF TBW	8.0
Flyback	No
QuickSat	1
Excitation Orientation	SS - Slice Selective
Ernst Angle T1	1220 ms
Reduce Slab Thickness?	Weak
skipped-CAIPI?	Off
Log Signals	Off

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\Physics_playground\Executed_20260603_Skopelwip_3DEPI_seg_fid_E
S1

TA: 7 sec Coil Selection: Manual Voxel Size: 0.9×0.9×0.9 mm³ Acc:: 3 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	-50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	30
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	190 mm
FOV Phase	100.0 %
Slice Thickness	0.88 mm
TR	50.0 ms
TE	19.00 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	50.0 ms
TE	19.00 ms
MTC	Off
Flip Angle	10 deg
Fat-Water Contrast	Fat Saturation
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off
3D Reordering	Standard

Contrast - Dynamic

Reordering	Linear
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Resolution - Common

FOV Read	190 mm
FOV Phase	100.0 %
Slice Thickness	0.88 mm
Base Resolution	208
Phase Resolution	98 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Reference Scans	GRE/Separate
Acceleration Factor PE	1
Reference Lines PE	24
Acceleration Factor 3D	3
Reference Lines 3D	24
Reordering Shift 3D	1
Deep Resolve	Off
Phase Partial Fourier	Off
Slice Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	30
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	190 mm
FOV Phase	100.0 %
Slice Thickness	0.88 mm
TR	50.0 ms
Multi-Slice Mode	Sequential
Series	Ascending

Geometry - Common

Concatenations	1
----------------	---

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.135464 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	50.0 ms
Segments	3
Concatenations	1

Sequence - Part 1

Sequence Name	WIP_epi
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast*
Flow Compensation	None
Reordering	Linear
Bandwidth	1202 Hz/Px
Echo Spacing	1.00 ms
Free Echo Spacing	Off
Segments	3
EPI Factor	25

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Correction	Internal

Sequence - Special

Prep/dummy scan duration	2 sec
Selective RF Duration	1024 us
Desired RF TBW	8.0
Flyback	No
QuickSat	1
Excitation Orientation	SS - Slice Selective
Ernst Angle T1	1220 ms
Reduce Slab Thickness?	Weak
skipped-CAIPI?	Off
Log Signals	Off

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\Physics_playground\Executed_20260603_Skope\vaso_2p3mm_single
_polarity_singloe_shot

TA: 43 sec Coil Selection: Manual Voxel Size: 2.3×2.3×2.3 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	40
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.30 mm
TR	47.4 ms
Vol. TR	4115.2 ms
TE 1	15.70 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	47.4 ms
Vol. TR	4115.2 ms
TE 1	15.70 ms
Multi-echo spacing	42.40 ms
MTC	Off
Magn. Preparation	Non-sel. IR
TI 1	1259.60 ms
TI 2	3155.60 ms
Flip Angle	18 deg
Fat-Water Contrast	Fast Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	10
Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.30 mm
Base Resolution	92
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	40
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.30 mm
TR	47.4 ms
Vol. TR	4115.2 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 H0.0
L	0.0 mm

Geometry - AutoAlign

A	20.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Quick
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.6 A6.7 F0.3 mm
! Orientation	T > 50.4 > C-0.1
! Rotation	90.14 deg
! R >> L	123 mm
! A >> P	169 mm
! F >> H	107 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.135464 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	0.200

Sequence - Part 1

Sequence Name	vso vaso_v6-dev
Dimension	3D

Sequence - Part 1

Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1874 Hz/Px
Echo Spacing	0.60 ms
Turbo Factor	40
Segmentation	1
EPI Factor	70

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	1000 us
RF time x BW	15
PAT ref. FA	3 deg
Fat sat. FA	110 deg
Variable FA	4
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Ramp Sampling	On
Trigger per shot	On
Noise image	Off
Relax spoilers	Off
Round up Vol. TR	Off
MT flip angle	500 deg
MT off-res.	2000 Hz
MT RF duration	10240 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI rise time factor	1.20
G. spoil dephasing[1]	0.0 pi
G. spoil dephasing[2]	4.0 pi
G. spoil dephasing[3]	2.0 pi
Modify Ice Config	On
G-factor map	Off
GRAPPA Regularization	10 /10^6
Slab Scale	-10 %
RF spoil scheme	Conventional
Read polarity	Regular RO
Pre-T11 delay	300.00 ms
Pre-T12 delay	0.00 ms
Pre-Inv delay	0.00 ms

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\Physics_playground\Executed_20260603_Skope\vaso_2p3mm_alternating_polarity

TA: 43 sec Coil Selection: Manual Voxel Size: 2.3×2.3×2.3 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	40
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.30 mm
TR	47.4 ms
Vol. TR	4115.2 ms
TE 1	15.70 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	47.4 ms
Vol. TR	4115.2 ms
TE 1	15.70 ms
Multi-echo spacing	42.40 ms
MTC	Off
Magn. Preparation	Non-sel. IR
TI 1	1259.60 ms
TI 2	3155.60 ms
Flip Angle	18 deg
Fat-Water Contrast	Fast Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	10
Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.30 mm
Base Resolution	92
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	40
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.30 mm
TR	47.4 ms
Vol. TR	4115.2 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 H0.0
L	0.0 mm

Geometry - AutoAlign

A	20.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Quick
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.6 A6.7 F0.3 mm
! Orientation	T > 50.4 > C-0.1
! Rotation	90.14 deg
! R >> L	123 mm
! A >> P	169 mm
! F >> H	107 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.135464 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	0.200

Sequence - Part 1

Sequence Name	vso vaso_v6-dev
Dimension	3D

Sequence - Part 1

Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1874 Hz/Px
Echo Spacing	0.60 ms
Turbo Factor	40
Segmentation	1
EPI Factor	70

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	1000 us
RF time x BW	15
PAT ref. FA	3 deg
Fat sat. FA	110 deg
Variable FA	4
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Ramp Sampling	On
Trigger per shot	On
Noise image	Off
Relax spoilers	Off
Round up Vol. TR	Off
MT flip angle	500 deg
MT off-res.	2000 Hz
MT RF duration	10240 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI rise time factor	1.20
G. spoil dephasing[1]	0.0 pi
G. spoil dephasing[2]	4.0 pi
G. spoil dephasing[3]	2.0 pi
Modify Ice Config	On
G-factor map	Off
GRAPPA Regularization	10 /10^6
Slab Scale	-10 %
RF spoil scheme	Conventional
Read polarity	Regular RO
Pre-T11 delay	300.00 ms
Pre-T12 delay	0.00 ms
Pre-Inv delay	0.00 ms

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\Physics_playground\Executed_20260603_Skopelvaso_seg4_noGRAP
PA_single_polarity

TA: 1:16 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	18
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	51.5 ms
Vol. TR	7439.2 ms
TE 1	19.00 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	51.5 ms
Vol. TR	7439.2 ms
TE 1	19.00 ms
Multi-echo spacing	40.20 ms
MTC	Off
Magn. Preparation	Non-sel. IR
TI 1	1917.10 ms
TI 2	5625.10 ms
Flip Angle	50 deg
Fat-Water Contrast	Fast Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	10
Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	206
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	18
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	51.5 ms
Vol. TR	7439.2 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 H0.0
L	0.0 mm

Geometry - AutoAlign

A	20.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Quick
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.6 A4.0 H0.6 mm
! Orientation	T > S0.4 > C-0.1
! Rotation	90.14 deg
! R >> L	123 mm
! A >> P	169 mm
! F >> H	42 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.135464 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	0.200

Sequence - Part 1

Sequence Name	vso vaso_v6-dev
Dimension	3D

Sequence - Part 1

Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1104 Hz/Px
Echo Spacing	1.01 ms
Turbo Factor	72
Segmentation	4
EPI Factor	39

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	1000 us
RF time x BW	15
PAT ref. FA	3 deg
Fat sat. FA	110 deg
Variable FA	4
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Ramp Sampling	On
Trigger per shot	On
Noise image	Off
Relax spoilers	Off
Round up Vol. TR	Off
MT flip angle	500 deg
MT off-res.	2000 Hz
MT RF duration	10240 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI rise time factor	1.20
G. spoil dephasing[1]	0.0 pi
G. spoil dephasing[2]	4.0 pi
G. spoil dephasing[3]	2.0 pi
Modify Ice Config	On
G-factor map	Off
GRAPPA Regularization	10 / 10^6
Slab Scale	-10 %
RF spoil scheme	Conventional
Read polarity	Regular RO
Pre-T11 delay	0.00 ms
Pre-T12 delay	0.00 ms
Pre-Inv delay	0.00 ms

Sequence - Assistant

SAR Assistant	Off
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MARTINOS DEVELOPER\HUBER\Physics_playground\Executed_20260603_Skopelvaso_seg1_GRAPPA_4_single_polarity

TA: 40 sec Coil Selection: Manual Voxel Size: 1.0x1.0x1.0 mm³ Acc:: 4 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	18
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	51.5 ms
Vol. TR	3377.2 ms
TE 1	19.00 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	51.5 ms
Vol. TR	3377.2 ms
TE 1	19.00 ms
Multi-echo spacing	40.20 ms
MTC	Off
Magn. Preparation	Non-sel. IR
TI 1	975.10 ms
TI 2	2402.10 ms
Flip Angle	50 deg
Fat-Water Contrast	Fast Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	10
Reordering	Linear

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	206
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	48
Acceleration Factor 3D	1
Reference Lines 3D	18
Reordering Shift 3D	0
Phase Partial Fourier	6/8
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	18
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	51.5 ms
Vol. TR	3377.2 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
------------	---

Geometry - AutoAlign

Position	L0.0 A20.0 H0.0 mm
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 H0.0
L	0.0 mm
A	20.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Quick
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.6 A4.0 H0.6 mm
! Orientation	T > S0.4 > C-0.1
! Rotation	90.14 deg
! R >> L	123 mm
! A >> P	169 mm
! F >> H	42 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.135464 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off

System - Tx/Rx

Correction Factor	1.00
Image Scaling	0.200

Sequence - Part 1

Sequence Name	vso vaso_v6-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1104 Hz/Px
Echo Spacing	1.01 ms
Turbo Factor	18
Segmentation	1
EPI Factor	39

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	1000 us
RF time x BW	15
PAT ref. FA	3 deg
Fat sat. FA	110 deg
Variable FA	4
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Ramp Sampling	On
Trigger per shot	On
Noise image	Off
Relax spoilers	Off
Round up Vol. TR	Off
MT flip angle	500 deg
MT off-res.	2000 Hz
MT RF duration	10240 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI rise time factor	1.20
G. spoil dephasing[1]	0.0 pi
G. spoil dephasing[2]	4.0 pi
G. spoil dephasing[3]	2.0 pi
Modify Ice Config	On
G-factor map	Off
GRAPPA Regularization	10 / 10^6
Slab Scale	-10 %
RF spoil scheme	Conventional
Read polarity	Regular RO
Pre-T11 delay	500.00 ms
Pre-T12 delay	500.00 ms
Pre-Inv delay	500.00 ms

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\Physics_playground\Executed_20260603_Skopelwip_3DEPI_seg_fid_E
S1

TA: 7 sec Coil Selection: Manual Voxel Size: 0.9×0.9×0.9 mm³ Acc:: 3 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	-50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	30
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	190 mm
FOV Phase	100.0 %
Slice Thickness	0.88 mm
TR	50.0 ms
TE	19.00 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	50.0 ms
TE	19.00 ms
MTC	Off
Flip Angle	10 deg
Fat-Water Contrast	Fat Saturation
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off
3D Reordering	Standard

Contrast - Dynamic

Reordering	Linear
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Resolution - Common

FOV Read	190 mm
FOV Phase	100.0 %
Slice Thickness	0.88 mm
Base Resolution	208
Phase Resolution	98 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Reference Scans	GRE/Separate
Acceleration Factor PE	1
Reference Lines PE	24
Acceleration Factor 3D	3
Reference Lines 3D	24
Reordering Shift 3D	1
Deep Resolve	Off
Phase Partial Fourier	Off
Slice Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	30
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	190 mm
FOV Phase	100.0 %
Slice Thickness	0.88 mm
TR	50.0 ms
Multi-Slice Mode	Sequential
Series	Ascending

Geometry - Common

Concatenations	1
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Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.135464 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	50.0 ms
Segments	3
Concatenations	1

Sequence - Part 1

Sequence Name	WIP_epi
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast*
Flow Compensation	None
Reordering	Linear
Bandwidth	1202 Hz/Px
Echo Spacing	1.00 ms
Free Echo Spacing	Off
Segments	3
EPI Factor	25

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Correction	Internal

Sequence - Special

Prep/dummy scan duration	2 sec
Selective RF Duration	1024 us
Desired RF TBW	8.0
Flyback	No
QuickSat	1
Excitation Orientation	SS - Slice Selective
Ernst Angle T1	1220 ms
Reduce Slab Thickness?	Weak
skipped-CAIPI?	Off
Log Signals	Off

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\Physics_playground\Executed_20260603_Skopelwip_3DEPI_seg_fid_E
S1_4_Nicolas

TA: 7 sec Coil Selection: Manual Voxel Size: 0.9×0.9×0.9 mm³ Acc:: 3 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	-50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	30
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	190 mm
FOV Phase	100.0 %
Slice Thickness	0.88 mm
TR	50.0 ms
TE	19.00 ms
Averages	1
Concatenations	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	50.0 ms
TE	19.00 ms
MTC	Off
Flip Angle	10 deg
Fat-Water Contrast	Fat Saturation
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off
3D Reordering	Standard

Contrast - Dynamic

Reordering	Linear
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Resolution - Common

FOV Read	190 mm
FOV Phase	100.0 %
Slice Thickness	0.88 mm
Base Resolution	208
Phase Resolution	98 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Reference Scans	GRE/Separate
Acceleration Factor PE	1
Reference Lines PE	24
Acceleration Factor 3D	3
Reference Lines 3D	24
Reordering Shift 3D	1
Deep Resolve	Off
Phase Partial Fourier	Off
Slice Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	30
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	190 mm
FOV Phase	100.0 %
Slice Thickness	0.88 mm
TR	50.0 ms
Multi-Slice Mode	Sequential
Series	Ascending

Geometry - Common

Concatenations	1
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Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.135464 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	50.0 ms
Segments	3
Concatenations	1

Sequence - Part 1

Sequence Name	WIP_epi
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast*
Flow Compensation	None
Reordering	Linear
Bandwidth	1202 Hz/Px
Echo Spacing	1.00 ms
Free Echo Spacing	Off
Segments	3
EPI Factor	25

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Correction	Internal

Sequence - Special

Prep/dummy scan duration	2 sec
Selective RF Duration	1024 us
Desired RF TBW	8.0
Flyback	No
QuickSat	1
Excitation Orientation	SS - Slice Selective
Ernst Angle T1	1220 ms
Reduce Slab Thickness?	Weak
skipped-CAIPI?	Off
Log Signals	Off

Sequence - Assistant

SAR Assistant	Off
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